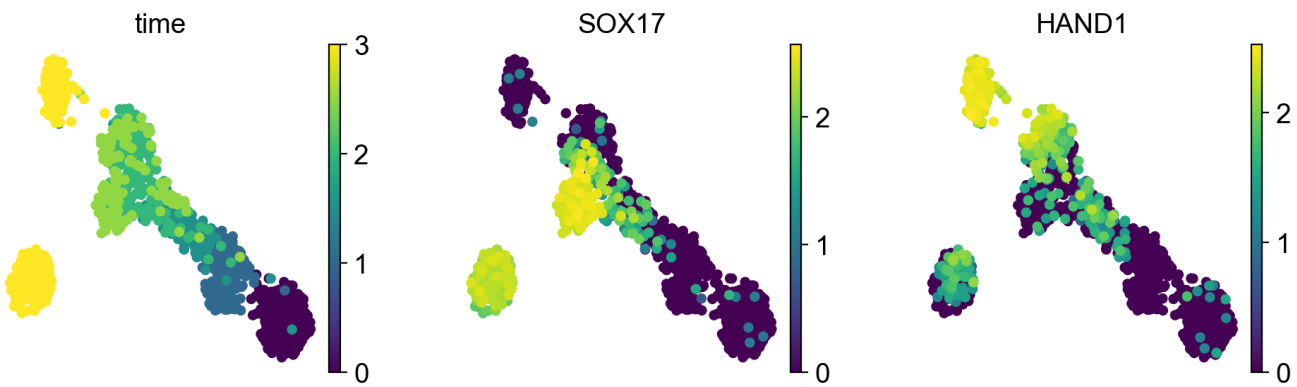
MuTrans Analaysis of IPS Reprogramming Dataset ¶

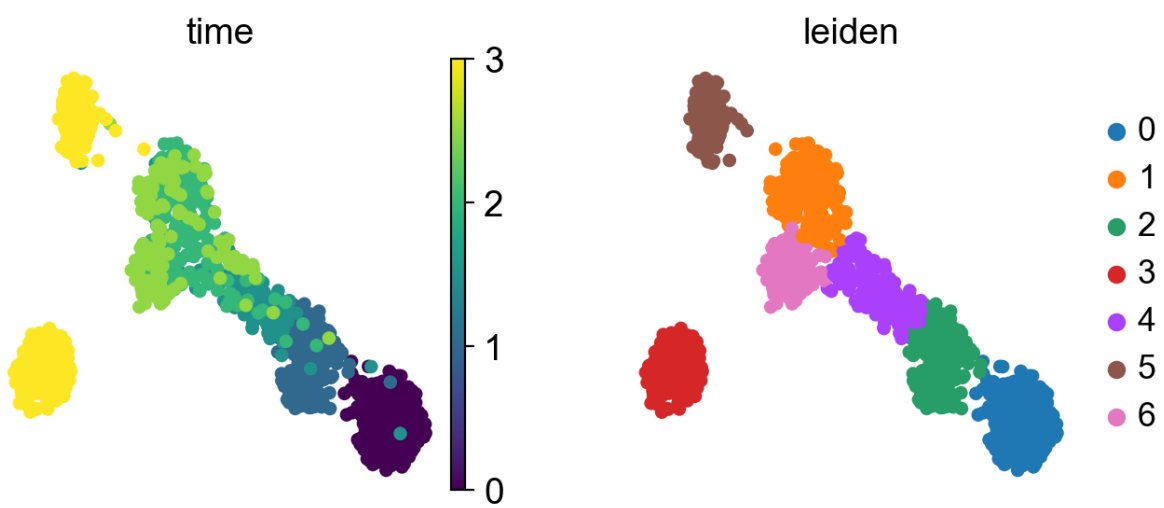
In [1]:
```
import pandas as pd
import numpy as np
import scanpy as sc
sc.settings.set_figure_params(dpi=100, frameon=False, figsize=(3, 3), facecolor='whit
e')
import matplotlib.pyplot as plt
import pyMuTrans as pm

datadir = "../Data/"
adata = sc.read_csv(datadir+'data_ipsc.csv') # load the pre-processed data
genes = pd.read_csv(datadir+'ipsc_genes.csv',header=None)
time = pd.read_csv(datadir+'ipsc_time.csv',header= None)
adata.var_names = genes.iloc[:,0]
adata.obs['time'] = time.values
```

In [2]:
```
sc.pp.neighbors(adata,metric = 'cosine',n_neighbors=60, use_rep='X')
sc.tl.umap(adata)
sc.pl.umap(adata, color = ['time','SOX17','HAND1'])
```

```
/Users/cliffzhou/opt/anaconda3/lib/python3.7/site-packages/pyemma/__init__.py:92: Us
erWarning: You are not using the latest release of PyEMMA. Latest is 2.5.7, you have
2.5.6.
  .format(latest=latest, current=current), category=UserWarning)
```

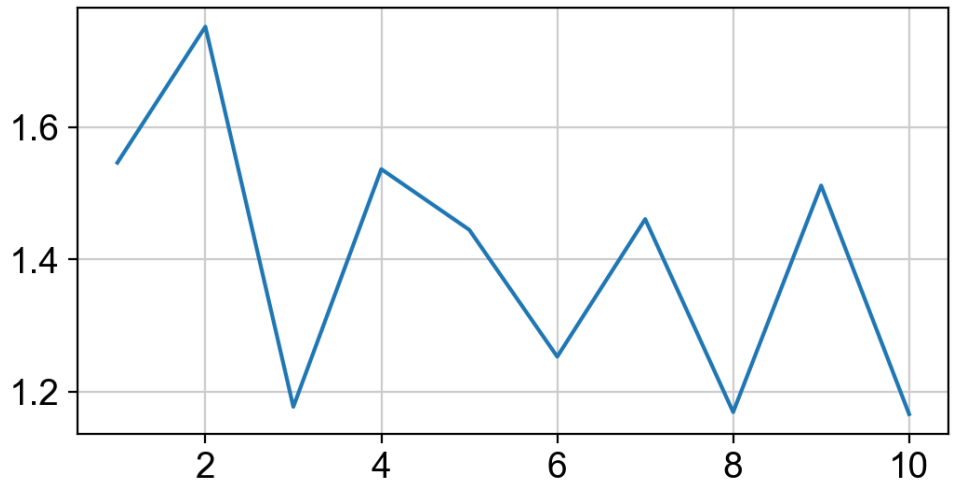


In [5]:
```
sc.tl.leiden(adata,resolution = 1.0)
sc.pl.umap(adata, color=['time','leiden'])
```



Estimate Cluster Number with EPI (Eigen-Peak Index)

In [6]:
```
fig = plt.figure(figsize=(6, 3))
par = {"choice_distance":"euclid"}
out = pm.plot_cluster_num(adata, par, k_plot= 10) #look for the peaks
```

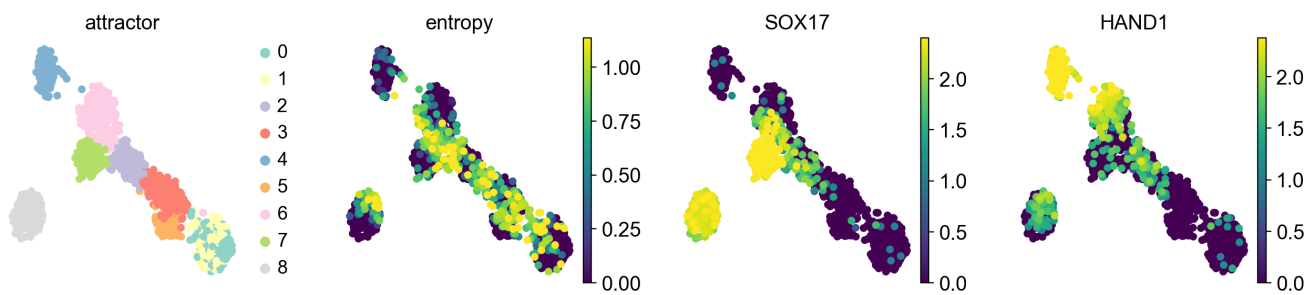


MuTrans Analysis Based on the AnnData Object

In [7]:
```
par = {"choice_distance":"cosine","perplex":200.0,"K_cluster":9.0,"trials":10,"reduct
ion_coord":'umap',"weight_scale":False,"write_anndata":True}
adata = pm.dynamical_analysis(adata,par)
```

In [8]:
```
import seaborn as sns
color_palette = sns.color_palette('Set3', 9)
temp = color_palette [6]
color_palette[6] = color_palette[7] # for reproduce the order in Matlab notebook
color_palette[7] = temp
sc.pl.umap(adata, color=['attractor','entropy','SOX17','HAND1'],vmax = 'p95',palette
= color_palette)
```

```
... storing 'attractor' as categorical
```



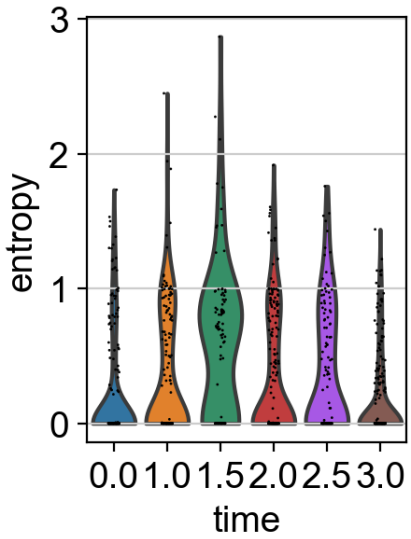
In [9]:
```
adata.obs['attractor'].value_counts()
```

Out[9]:
```
6    192
3    151
8    150
1    137
4    110
0     98
7     95
2     87
5     61
Name: attractor, dtype: int64
```

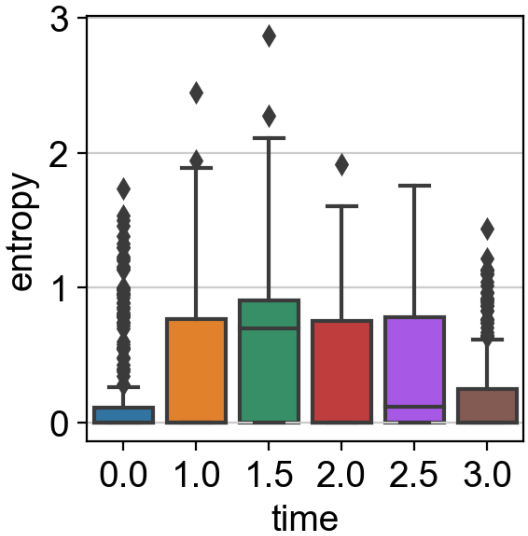
Examine the Entropy of Transition Cells

In [10]:
```
adata.obs['time'] = adata.obs['time'].astype(str)
sc.pl.violin(adata, keys = ['entropy'], groupby='time')
```

```
... storing 'time' as categorical
/Users/cliffzhou/opt/anaconda3/lib/python3.7/site-packages/seaborn/_decorators.py:4
3: FutureWarning: Pass the following variable as a keyword arg: x. From version 0.1
2, the only valid positional argument will be `data`, and passing other arguments wi
thout an explicit keyword will result in an error or misinterpretation.
  FutureWarning
/Users/cliffzhou/opt/anaconda3/lib/python3.7/site-packages/seaborn/_decorators.py:4
3: FutureWarning: Pass the following variable as a keyword arg: x. From version 0.1
2, the only valid positional argument will be `data`, and passing other arguments wi
thout an explicit keyword will result in an error or misinterpretation.
  FutureWarning
```



In [11]:
```
ax = sns.boxplot(x="time", y="entropy", data=adata.obs)
```



In [12]:
```
sc.pl.violin(adata, keys = ['entropy'], groupby='attractor',palette = color_palette)
```

```
/Users/cliffzhou/opt/anaconda3/lib/python3.7/site-packages/seaborn/_decorators.py:4
3: FutureWarning: Pass the following variable as a keyword arg: x. From version 0.1
2, the only valid positional argument will be `data`, and passing other arguments wi
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  FutureWarning
/Users/cliffzhou/opt/anaconda3/lib/python3.7/site-packages/seaborn/_decorators.py:4
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thout an explicit keyword will result in an error or misinterpretation.
  FutureWarning
```
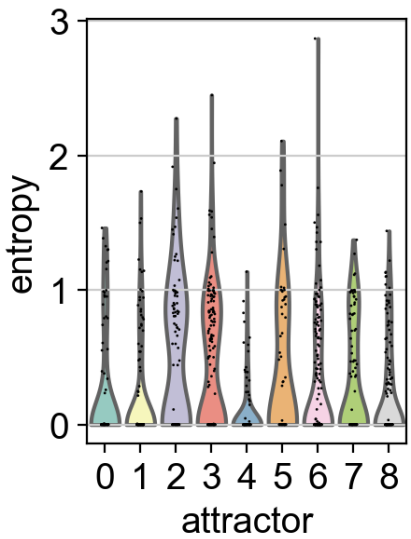
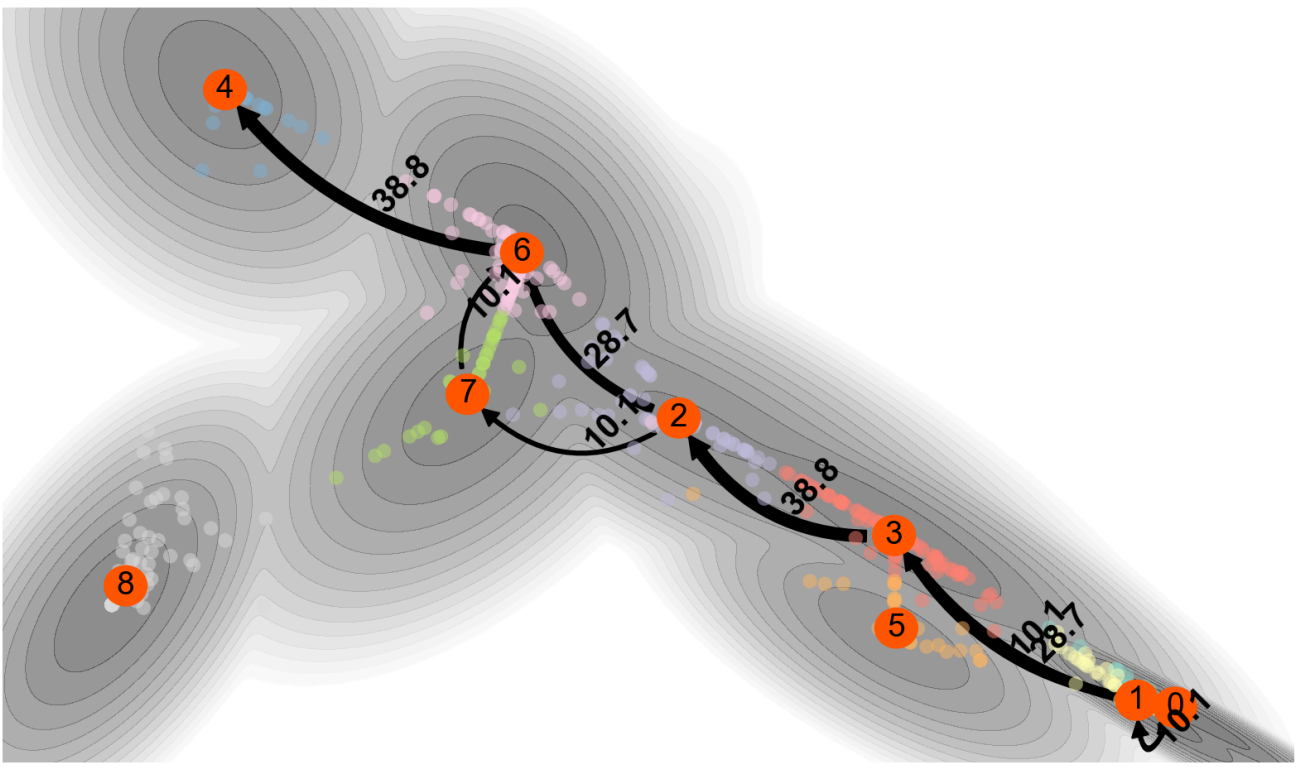
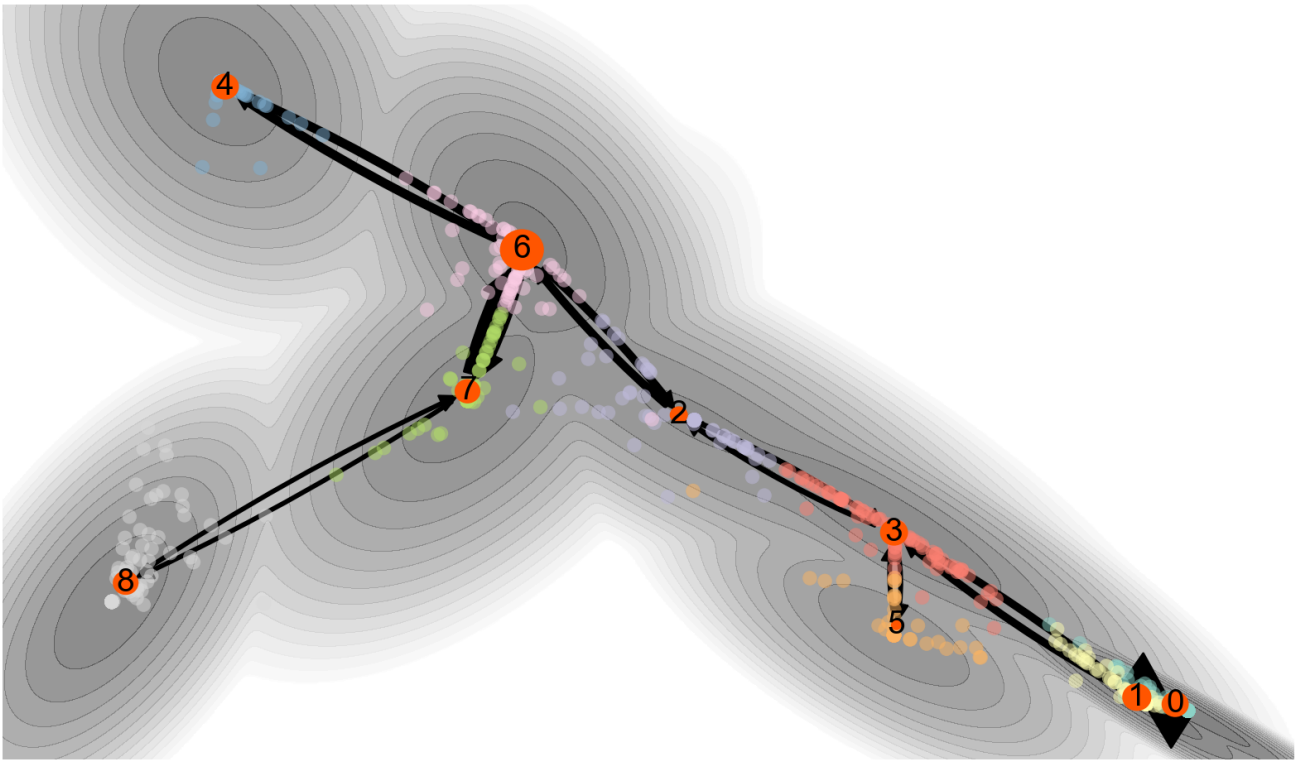



Transition Path Analysis

In [13]:
```
fig = plt.figure(figsize=(10, 6))
pm.infer_lineage(adata,si=0,sf=4,method = "MPPT",flux_fraction = 0.3,size_point =40,a
lpha_point = 0.5,size_text = 15,color_palette=color_palette)
```

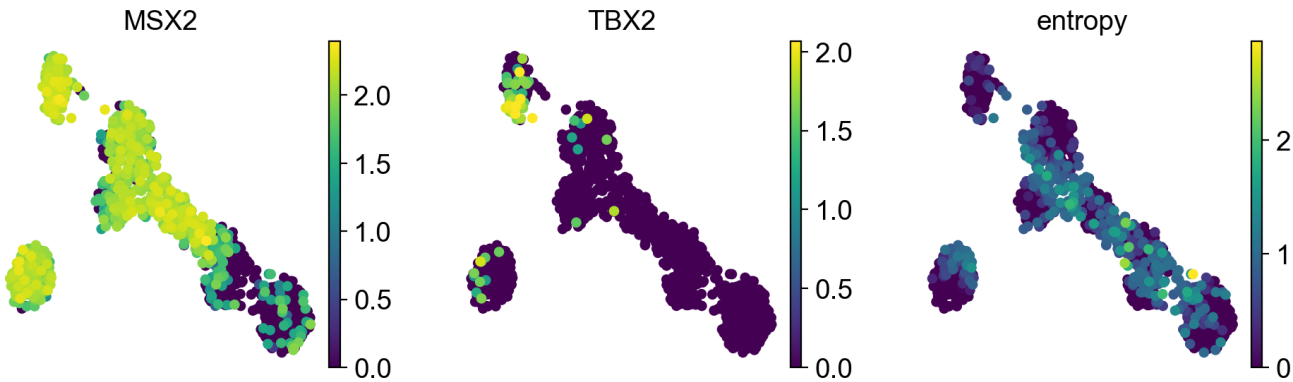


In [14]:
```
fig = plt.figure(figsize=(10, 6))
pm.infer_lineage(adata,si=0,sf=4,method = "MPFT",flux_fraction = 0.3,size_point =40,a
lpha_point = 0.5,size_text = 15,color_palette=color_palette)
```



Visualization of Gene Analysis Results

In [15]:
```
sc.pl.umap(adata, color=['MSX2','TBX2','entropy'])
```



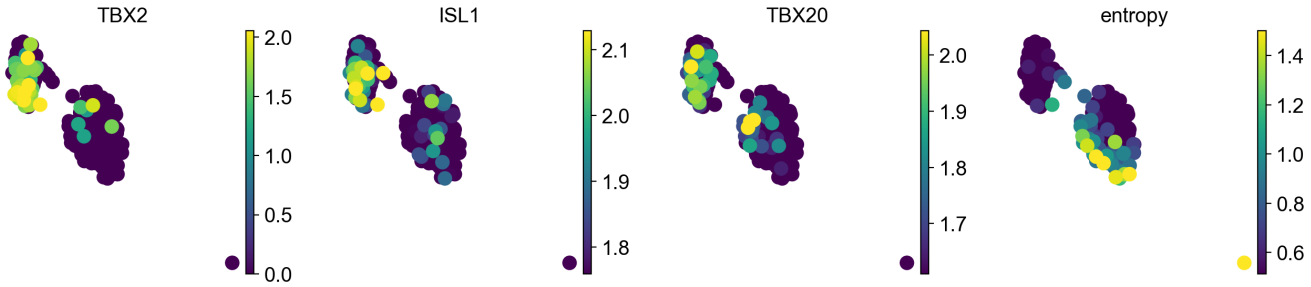
In [16]:
```
adata_sub = adata[adata.obs['attractor'].isin(['6', '4']),:]
```

```
/Users/cliffzhou/opt/anaconda3/lib/python3.7/site-packages/pandas/core/arrays/catego
rical.py:2487: FutureWarning: The `inplace` parameter in pandas.Categorical.remove_u
nused_categories is deprecated and will be removed in a future version.
  res = method(*args, **kwargs)
```

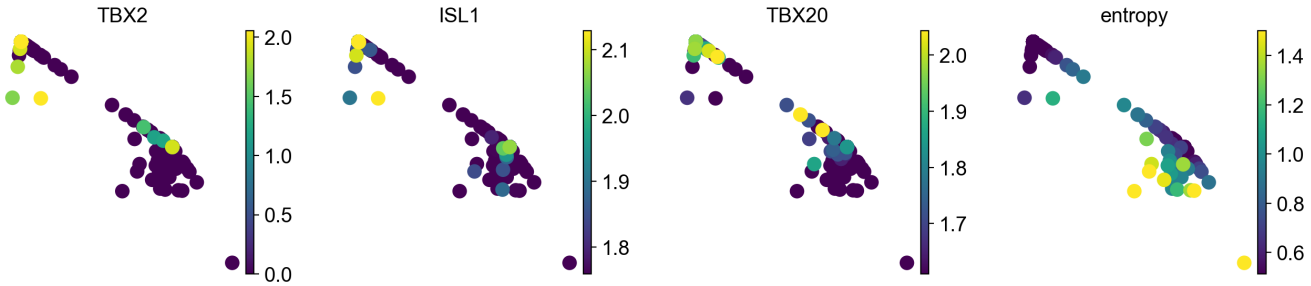
In [17]:
```
adata_sub
```

Out[17]:
```
View of AnnData object with n_obs × n_vars = 302 × 96
    obs: 'time', 'leiden', 'land', 'entropy', 'attractor'
    uns: 'neighbors', 'umap', 'leiden', 'leiden_colors', 'da_out', 'land', 'attracto
r_colors', 'time_colors'
    obsm: 'X_umap', 'trans_coord', 'membership'
    obsp: 'distances', 'connectivities'
```

In [18]:
```
sc.pl.umap(adata_sub, color=['TBX2','ISL1','TBX20','entropy'],vmin = 'p80',vmax = 'p9
9')
```



In [19]:
```
sc.pl.embedding(adata_sub, color=['TBX2','ISL1','TBX20','entropy'],vmin = 'p80',vmax
= 'p99',basis = 'trans_coord')
```



In [20]:
```
sc.pl.violin(adata_sub, keys = ['TBX2','TBX20','MYL4'], groupby='attractor',palette =
[color_palette[4],color_palette[6]])
```

```
/Users/cliffzhou/opt/anaconda3/lib/python3.7/site-packages/seaborn/_decorators.py:4
3: FutureWarning: Pass the following variable as a keyword arg: x. From version 0.1
2, the only valid positional argument will be `data`, and passing other arguments wi
thout an explicit keyword will result in an error or misinterpretation.
  FutureWarning
/Users/cliffzhou/opt/anaconda3/lib/python3.7/site-packages/seaborn/_decorators.py:4
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  FutureWarning
/Users/cliffzhou/opt/anaconda3/lib/python3.7/site-packages/seaborn/_decorators.py:4
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  FutureWarning
/Users/cliffzhou/opt/anaconda3/lib/python3.7/site-packages/seaborn/_decorators.py:4
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2, the only valid positional argument will be `data`, and passing other arguments wi
thout an explicit keyword will result in an error or misinterpretation.
  FutureWarning
```
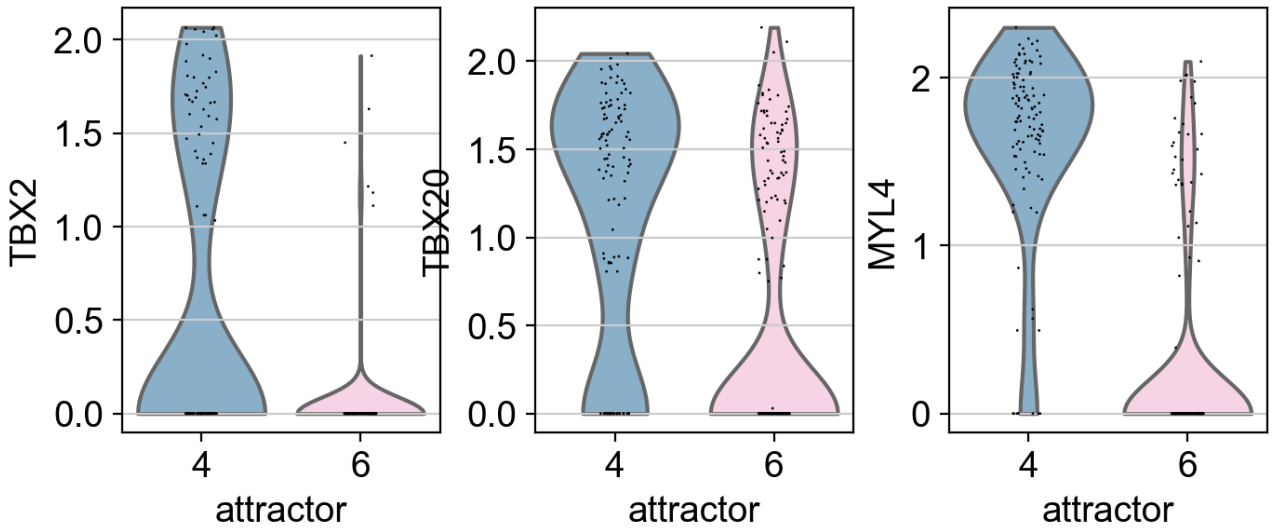
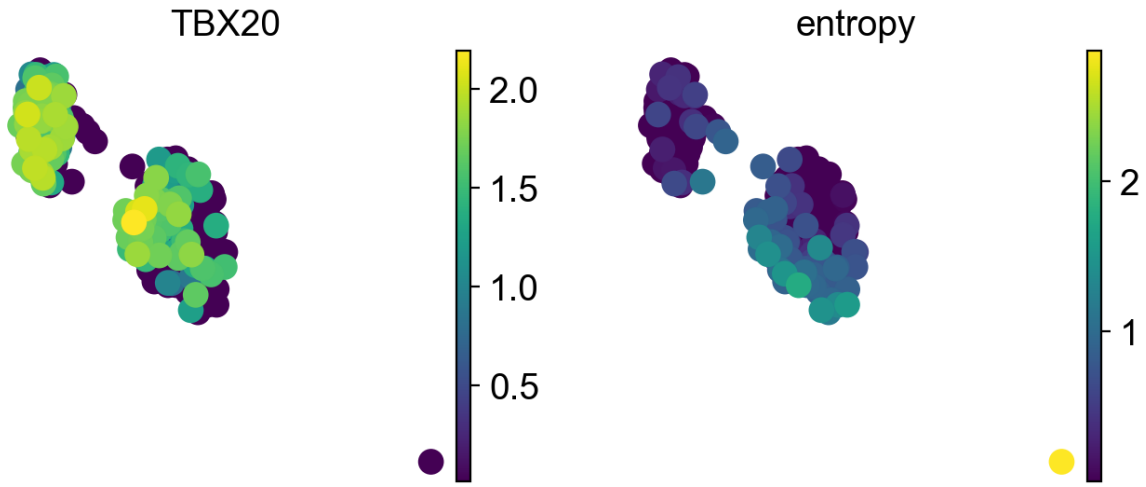
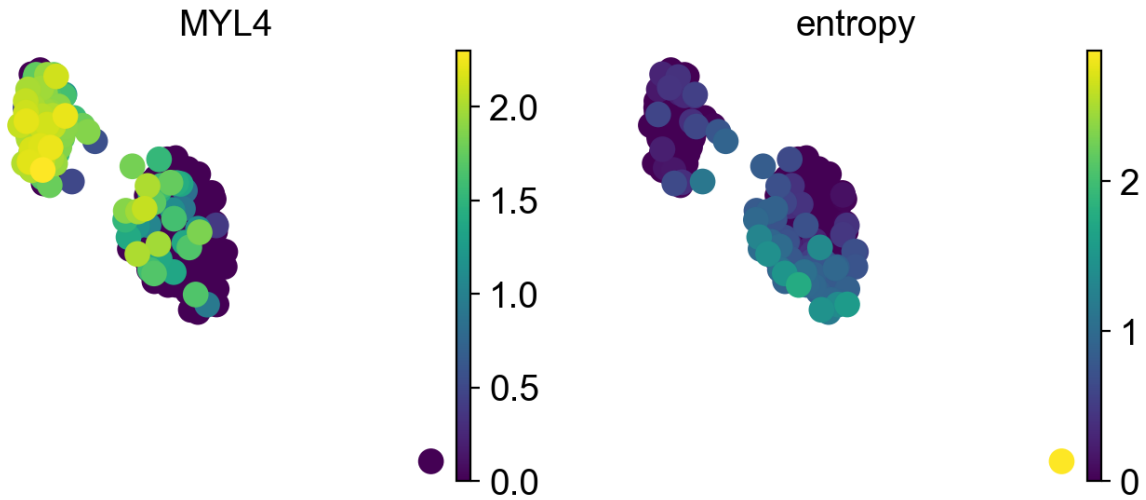



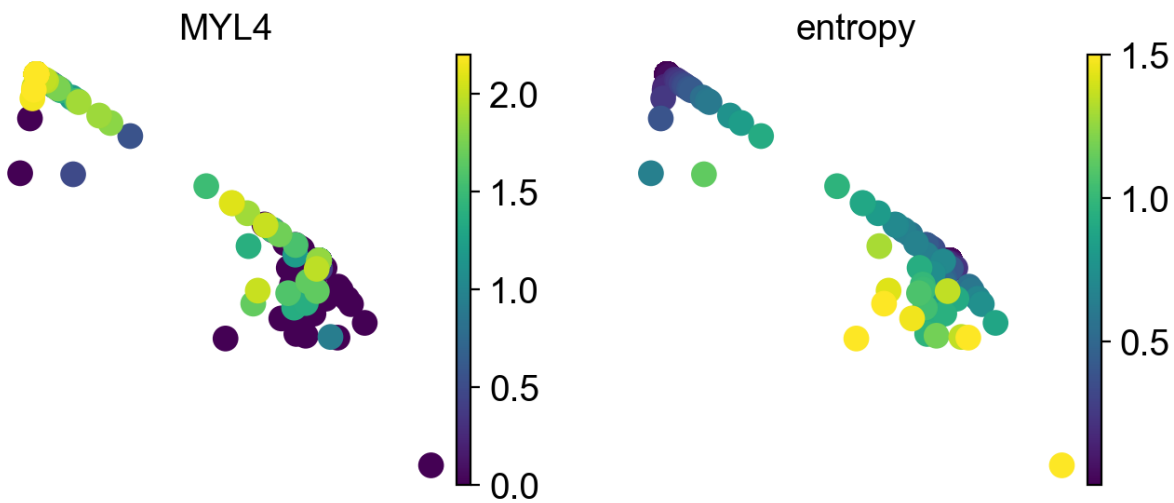
In [21]:
```
sc.pl.umap(adata_sub, color=['TBX20','entropy'],vmin = 'p50')
```



In [22]:
```
sc.pl.umap(adata_sub, color=['MYL4','entropy'])
```



In [23]:
```
sc.pl.embedding(adata_sub, color=['MYL4','entropy'],vmin = 'p20',vmax = 'p99',basis =
'trans_coord')
```

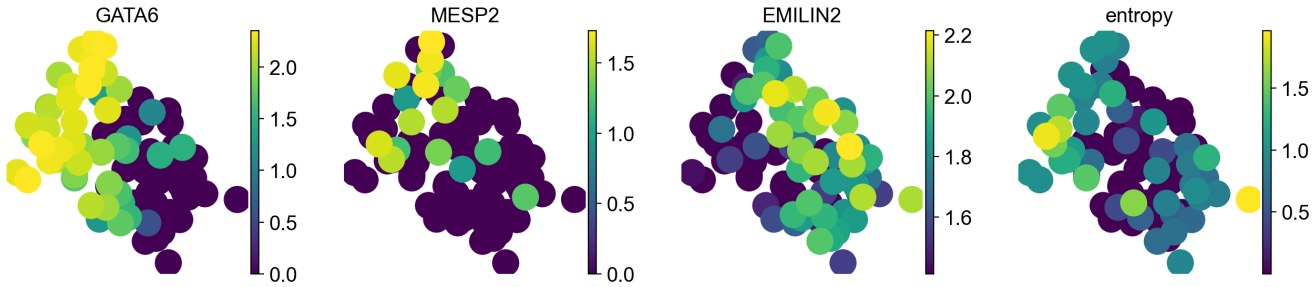


With the bifurcating state:

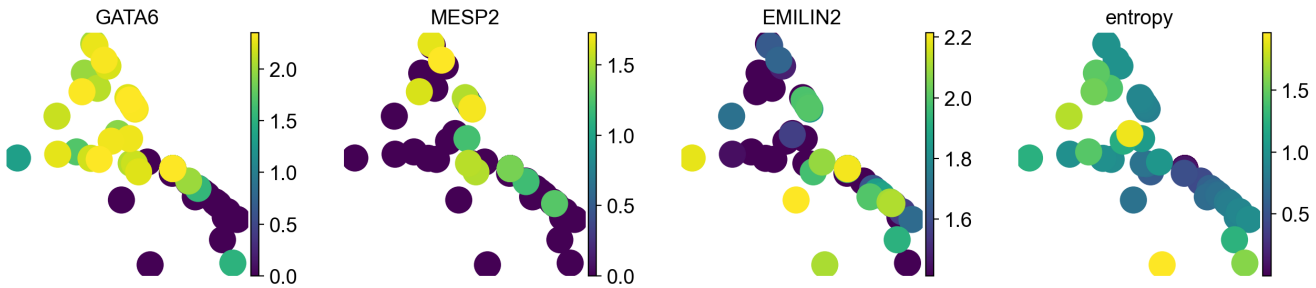
In [24]:
```
adata_ps = adata[adata.obs['attractor'].isin(['2']),:]
```

```
/Users/cliffzhou/opt/anaconda3/lib/python3.7/site-packages/pandas/core/arrays/catego
rical.py:2487: FutureWarning: The `inplace` parameter in pandas.Categorical.remove_u
nused_categories is deprecated and will be removed in a future version.
  res = method(*args, **kwargs)
```

In [25]:
```
sc.pl.umap(adata_ps, color=['GATA6','MESP2','EMILIN2','entropy'],vmin = 'p30',vmax =
'p99')
```



In [26]:
```
sc.pl.embedding(adata_ps, color=['GATA6','MESP2','EMILIN2','entropy'],vmin = 'p30',vm
ax = 'p99',basis = 'trans_coord')
```

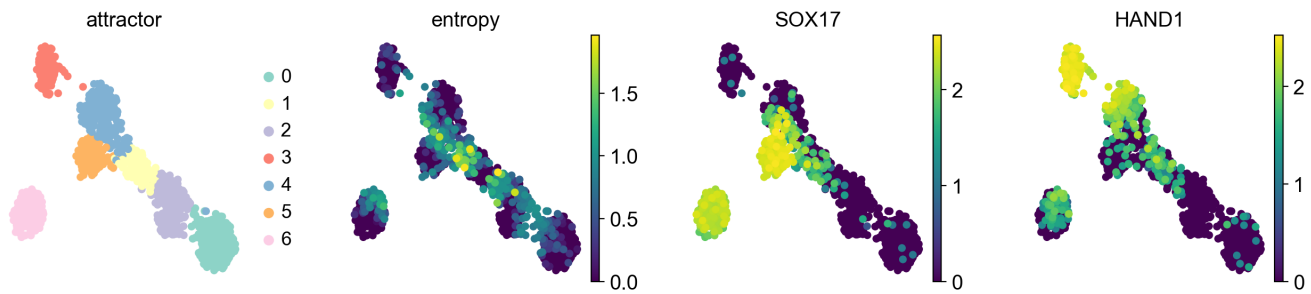


Changing the number of clusters K

In [27]:
```
par['K_cluster'] = 7.0
adata = pm.dynamical_analysis(adata,par)
```

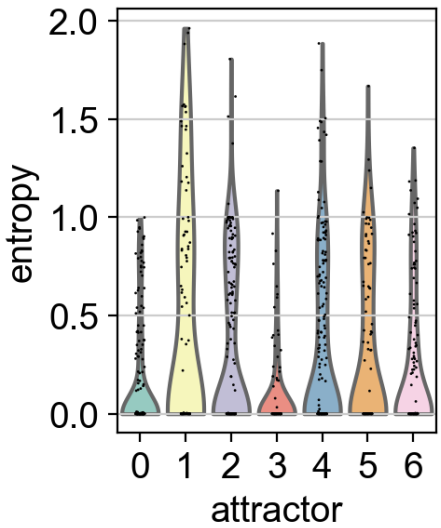
In [28]:
```
sc.pl.umap(adata, color=['attractor','entropy','SOX17','HAND1'],palette = color_palet
te)
```

```
... storing 'attractor' as categorical
```



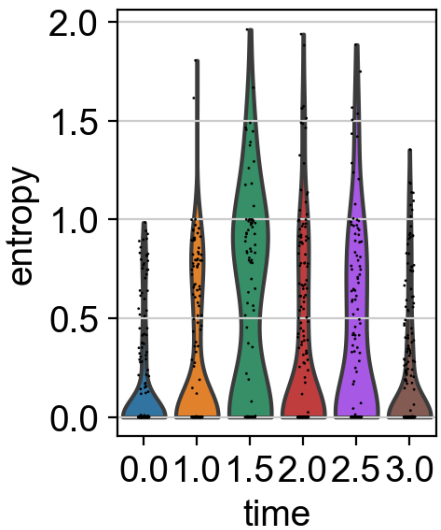
In [29]:
```
sc.pl.violin(adata, keys = ['entropy'], groupby='attractor')
```

```
/Users/cliffzhou/opt/anaconda3/lib/python3.7/site-packages/seaborn/_decorators.py:4
3: FutureWarning: Pass the following variable as a keyword arg: x. From version 0.1
2, the only valid positional argument will be `data`, and passing other arguments wi
thout an explicit keyword will result in an error or misinterpretation.
  FutureWarning
/Users/cliffzhou/opt/anaconda3/lib/python3.7/site-packages/seaborn/_decorators.py:4
3: FutureWarning: Pass the following variable as a keyword arg: x. From version 0.1
2, the only valid positional argument will be `data`, and passing other arguments wi
thout an explicit keyword will result in an error or misinterpretation.
  FutureWarning
```

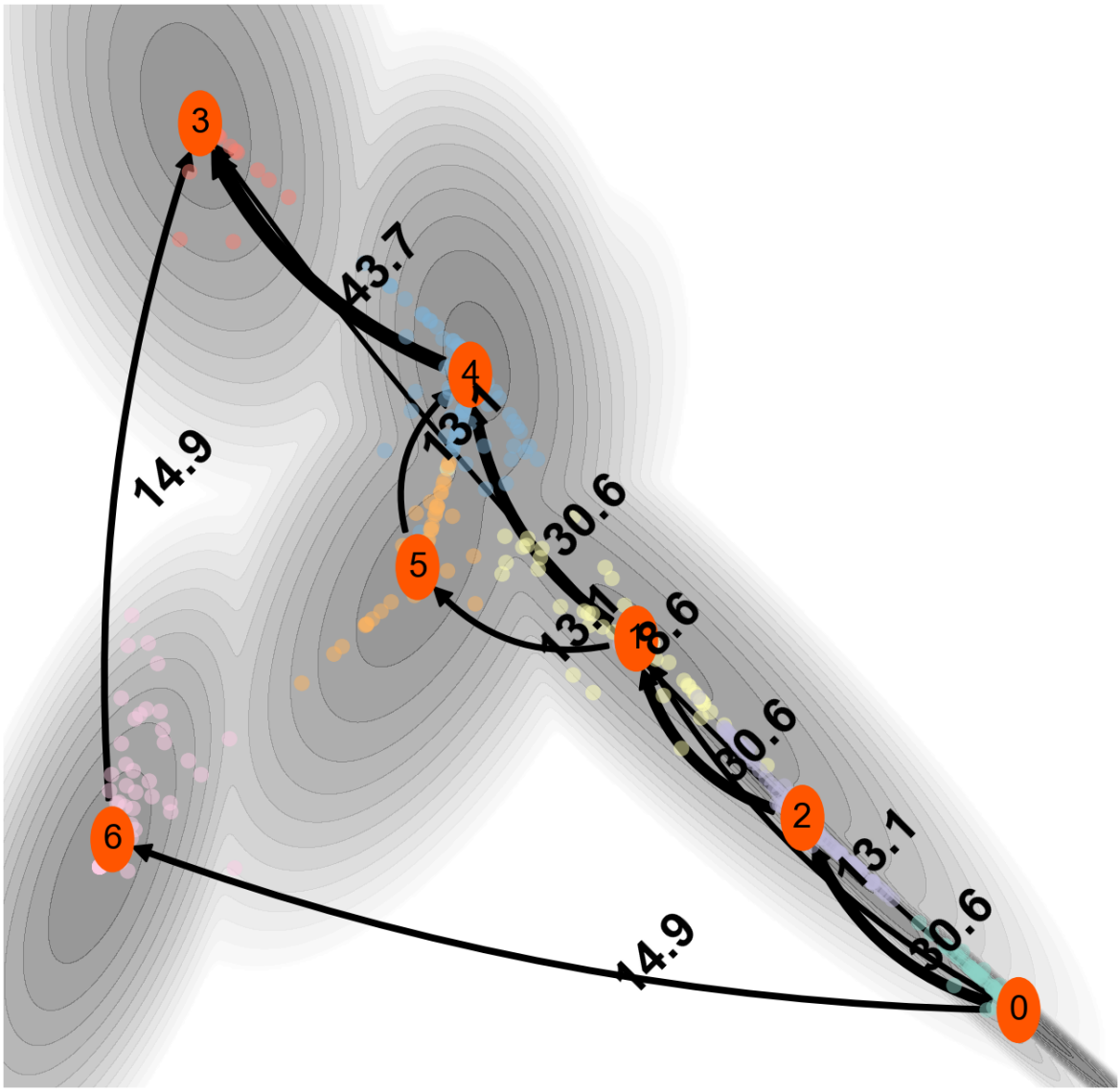


In [30]:
```
adata.obs['time'] = adata.obs['time'].astype(str)
sc.pl.violin(adata, keys = ['entropy'], groupby='time')
```

```
... storing 'time' as categorical
/Users/cliffzhou/opt/anaconda3/lib/python3.7/site-packages/seaborn/_decorators.py:4
3: FutureWarning: Pass the following variable as a keyword arg: x. From version 0.1
2, the only valid positional argument will be `data`, and passing other arguments wi
thout an explicit keyword will result in an error or misinterpretation.
  FutureWarning
/Users/cliffzhou/opt/anaconda3/lib/python3.7/site-packages/seaborn/_decorators.py:4
3: FutureWarning: Pass the following variable as a keyword arg: x. From version 0.1
2, the only valid positional argument will be `data`, and passing other arguments wi
thout an explicit keyword will result in an error or misinterpretation.
  FutureWarning
```

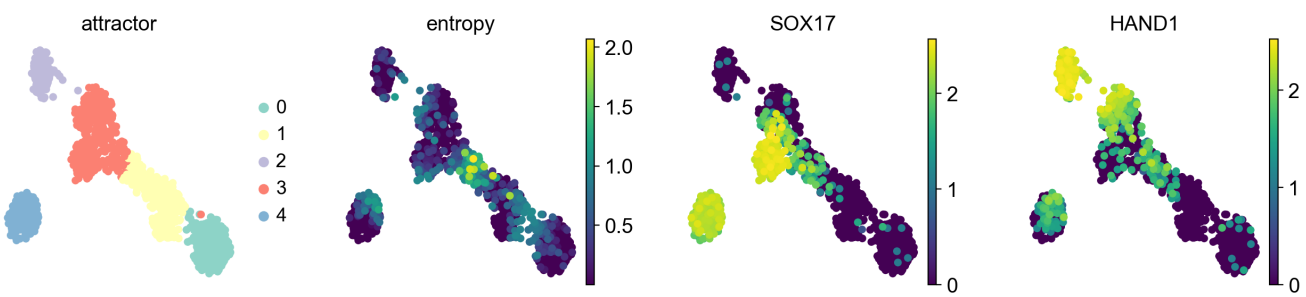


In [31]:
```
fig = plt.figure(figsize=(8, 8))
pm.infer_lineage(adata,si=0,sf=3,method = "MPPT",flux_fraction = 0.6,size_point =40,a
lpha_point = 0.5, color_palette =color_palette )
```



In [32]:
```
par['K_cluster'] = 5.0
adata = pm.dynamical_analysis(adata,par)
sc.pl.umap(adata, color=['attractor','entropy','SOX17','HAND1'])
```

```
... storing 'attractor' as categorical
```



In [33]:
```
adata.obs['time'] = adata.obs['time'].astype(str)
sc.pl.violin(adata, keys = ['entropy'], groupby='time')
```

```
... storing 'time' as categorical
/Users/cliffzhou/opt/anaconda3/lib/python3.7/site-packages/seaborn/_decorators.py:4
3: FutureWarning: Pass the following variable as a keyword arg: x. From version 0.1
2, the only valid positional argument will be `data`, and passing other arguments wi
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  FutureWarning
/Users/cliffzhou/opt/anaconda3/lib/python3.7/site-packages/seaborn/_decorators.py:4
3: FutureWarning: Pass the following variable as a keyword arg: x. From version 0.1
2, the only valid positional argument will be `data`, and passing other arguments wi
thout an explicit keyword will result in an error or misinterpretation.
  FutureWarning
```
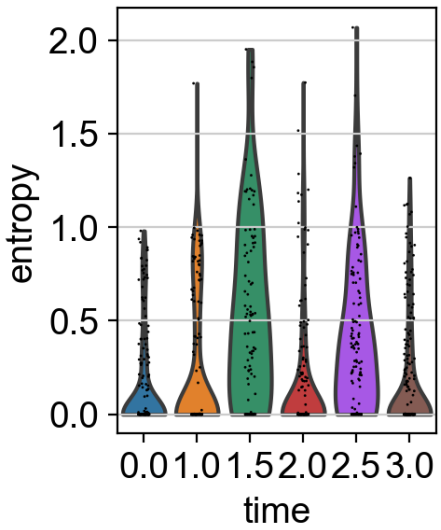
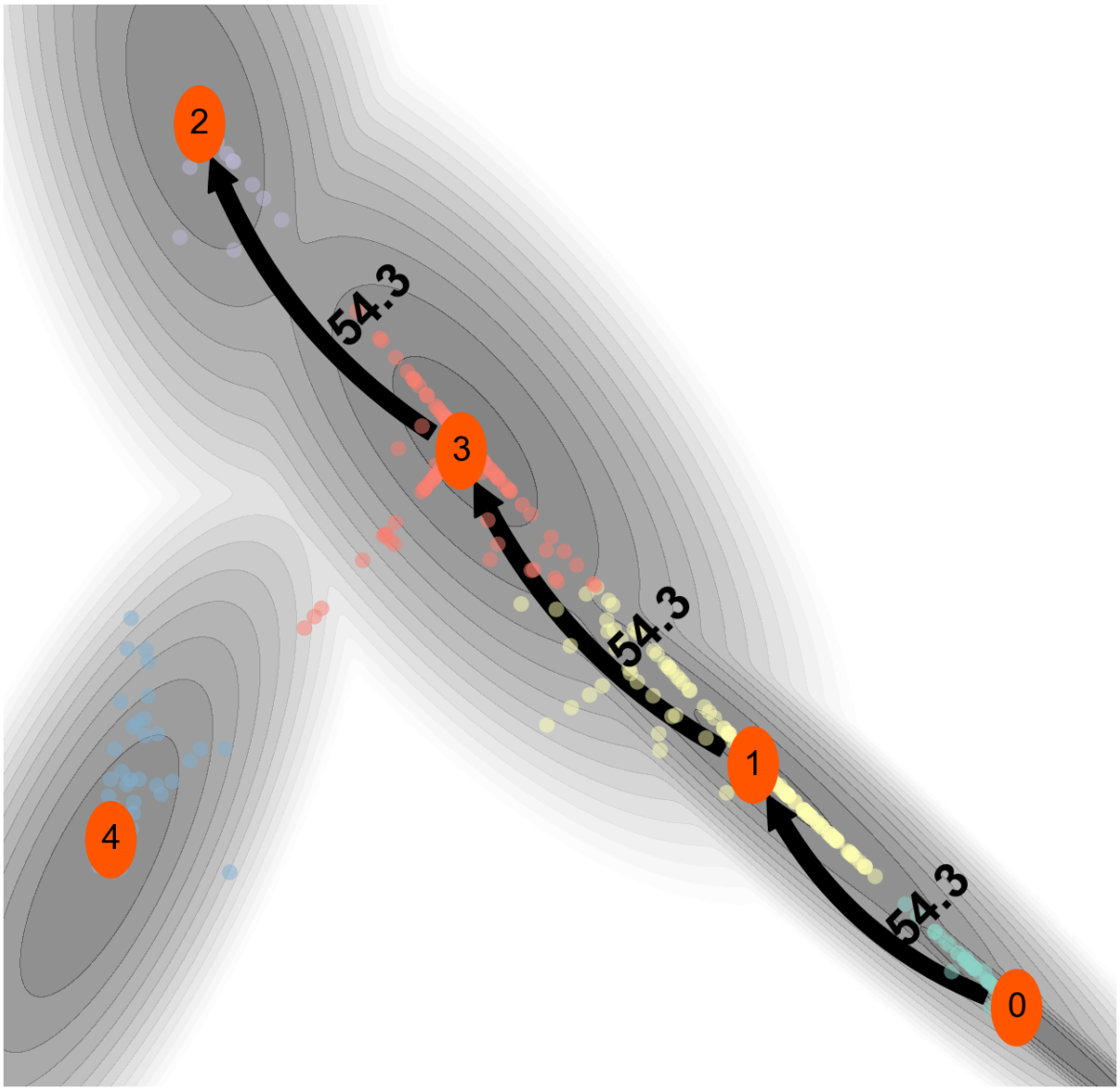



In [34]:
```
fig = plt.figure(figsize=(8, 8))
pm.infer_lineage(adata,si=0,sf=2,method = "MPPT",flux_fraction = 0.5,size_point =40,a
lpha_point = 0.5,color_palette =color_palette)
```



Population Balance Analysis

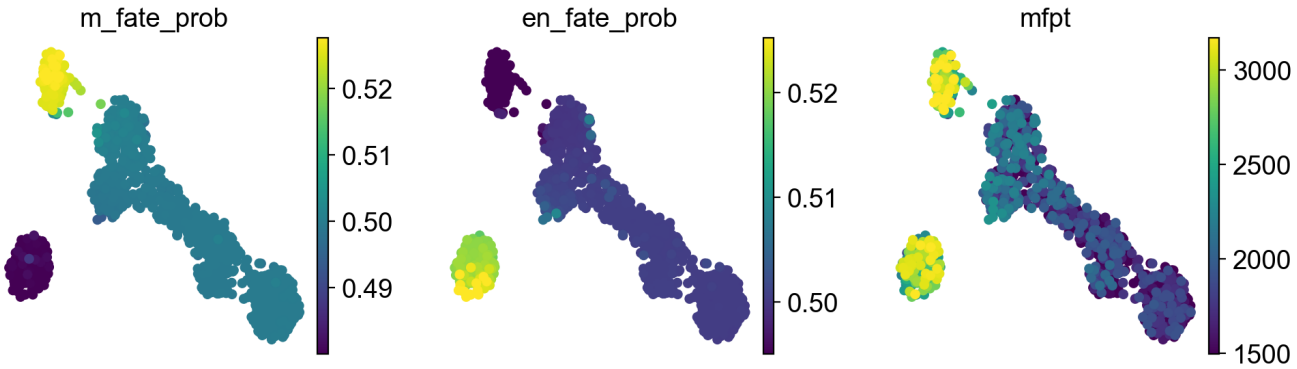
In [37]:
```
#output to PBA
import numpy as np
import random
np.save('X',adata.X)
R = np.zeros(1081)
np.save('R',R)
```

In [38]:
```
S = np.zeros((1081,2))
M_id = adata.obs['attractor'].to_numpy()=='2'
En_id = adata.obs['attractor'].to_numpy()=='4'
S[random.choice(np.argwhere(M_id == True)),0] = 1
S[random.choice(np.argwhere(En_id == True)),1] = 1
np.save('S',S)
```

In [39]:
```
# load the computation results of PBA
B = np.load('B_ipsc.npy')
V = np.load('V_ipsc.npy')
T = np.load('T_ipsc.npy')
```

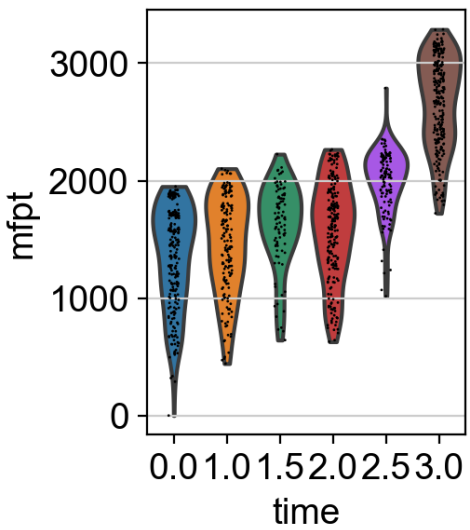
In [40]:
```
adata.obs['m_fate_prob'] = B[:,0]
adata.obs['en_fate_prob']= B[:,1]
np.random.seed(0)
init_cell = random.choice(np.argwhere(adata.obs['attractor'].to_numpy()=='0'))
adata.obs['mfpt'] = T[int(init_cell),:]
```

In [41]:
```
sc.pl.umap(adata, color=['m_fate_prob','en_fate_prob','mfpt'],vmin = ['p10','p10','p3
0'],vmax = 'p99')
```



In [42]:
```
sc.pl.violin(adata, keys = ['mfpt'], groupby='time')
```

```
/Users/cliffzhou/opt/anaconda3/lib/python3.7/site-packages/seaborn/_decorators.py:4
3: FutureWarning: Pass the following variable as a keyword arg: x. From version 0.1
2, the only valid positional argument will be `data`, and passing other arguments wi
thout an explicit keyword will result in an error or misinterpretation.
  FutureWarning
/Users/cliffzhou/opt/anaconda3/lib/python3.7/site-packages/seaborn/_decorators.py:4
3: FutureWarning: Pass the following variable as a keyword arg: x. From version 0.1
2, the only valid positional argument will be `data`, and passing other arguments wi
thout an explicit keyword will result in an error or misinterpretation.
  FutureWarning
```



In [43]:
```
from scipy.stats import kendalltau
kendalltau(adata.obs['time'],adata.obs['mfpt'])
```

Out[43]:
```
KendalltauResult(correlation=0.5650667643753641, pvalue=1.118139547464182e-144)
```

In [44]:
```
from scipy.stats import spearmanr
spearmanr(adata.obs['time'].to_numpy().astype('float'),adata.obs['mfpt'])
```

Out[44]:
```
SpearmanrResult(correlation=0.7257893638436187, pvalue=1.6809216414854869e-177)
```

In [45]:
```
from scipy.stats import pearsonr
pearsonr(adata.obs['time'].to_numpy().astype('float'),adata.obs['mfpt'])
```

Out[45]:
```
(0.6828735240410416, 2.6317022428179664e-149)
```